

1 The Energy Crossover Effect: How Model Architecture 2 Shapes Hardware Efficiency Across Generative AI 3 Modalities

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6 Abstract

Generative AI spans text, image, video, and music, yet how energy consumption scales across modalities and hardware remains poorly understood. We benchmark three hardware tiers—edge SoC (Apple M4 Pro, 22W), mainstream GPU (NVIDIA A100, 400W), and high-end GPU (NVIDIA H100, 700W)—across 4,500 runs, 6 models, and 76 configurations. We report an *Energy Crossover Effect*: for diffusion workloads, energy efficiency shifts from edge to GPU as output complexity grows, while autoregressive workloads favour edge hardware by 6–10× regardless of complexity. The crossover threshold depends on GPU generation—the H100 reaches parity between 256^2 and 384^2 pixels (vs. $\sim 466^2$ for A100) and is cheaper than edge for video at *all* tested sizes. Power-law fits ($E = \alpha \cdot N^\beta$) reveal superlinear edge scaling ($\beta > 1$) versus sublinear GPU scaling ($\beta < 1$) for diffusion, producing predictable crossover points absent in autoregressive tasks. These measurements yield deployment guidelines for energy-aware AI routing.

7 *Keywords:* Generative AI, Energy efficiency, Edge computing, GPU
8 benchmarking, Inference energy, Scaling laws, Sustainable AI

9 1. Introduction

10 Generative AI is now among the most energy-intensive workloads in com-
11 puting. Applications now span text generation (GPT-4, LLaMA, Phi-3 [1]),
12 image synthesis (Stable Diffusion [2], DALL-E 3), video creation (Sora, An-
13 imateDiff [3]), and music composition (MusicGen [4], MusicLM). The Inter-
14 national Energy Agency (IEA) projects that global data centre electricity

consumption will reach 945 TWh by 2030, with AI workloads as a primary driver [5]. A single ChatGPT query consumes approximately $10\times$ the energy of a Google search [6], and image generation uses over $60\times$ more energy than text classification per inference [7].

The starting point for our work is a simple observation: generative models use two very different computational paradigms. *Autoregressive* models (text, music) emit tokens one at a time, producing inherently serial computation. *Diffusion* models [2] (image, video) denoise all spatial positions in parallel, producing computation that scales with resolution. These paradigms should interact differently with hardware designed for throughput (data centre GPUs) versus hardware designed for efficiency (edge SoCs with unified memory and low power envelopes) [8]—and the interaction need not be monotonic: a GPU’s advantage may emerge only once the workload is large enough to keep its cores busy. At the same time, edge-capable AI hardware—Apple’s M-series, Qualcomm’s Snapdragon X Elite—has grown powerful enough to run billion-parameter models locally, making the edge-versus-cloud decision a live question for a broad class of workloads.

We frame the analysis in terms of scaling laws—power-law relationships between computational resources and behaviour—which have become a standard lens in AI research [9]. Existing scaling laws describe model performance and training cost; we fit analogous relationships to *inference energy*, and find that the exponents differ qualitatively across hardware. The practical stakes are large. Routing a text query to a data centre GPU wastes $6\text{--}10\times$ more energy than processing it on an edge device, while routing a high-resolution image to an edge device can be equally wasteful. With AI inference expanding to billions of daily queries [6], even small per-query savings compound to meaningful reductions in aggregate consumption. And because the relationship between energy and task complexity is not simply linear—different hardware platforms follow different scaling regimes—there are predictable crossover points where the optimal hardware choice switches.

Here we report the first cross-hardware, cross-modality energy benchmark spanning three hardware tiers, comprising 4,500 runs across 6 models and 76 configurations. The main contributions are: (1) identification and quantification of the **Energy Crossover Effect**—the shift in energy-efficiency leadership from edge to GPU as task complexity grows, occurring only in parallelisable (diffusion) workloads; (2) **modality-specific energy scaling laws** ($E = \alpha \cdot N^\beta$) showing superlinear edge scaling ($\beta > 1$) versus sublinear GPU scaling ($\beta < 1$) for diffusion; (3) evidence that **GPU generation**

53 **shifts the crossover threshold:** the H100 reaches parity with the Mac
54 between 256^2 and 384^2 pixels (vs. $\sim 466^2$ for A100) and eliminates the video
55 crossover altogether; and (4) **deployment guidelines** derived from these
56 measurements for energy-aware routing.

57 2. Related Work

58 2.1. Energy consumption of AI training and inference

59 Early work by Strubell et al. [10] brought attention to the energy cost
60 of training Transformer [11] models, and Patterson et al. [12] refined these
61 estimates for large-scale training. More recently, the focus has shifted to
62 inference: Luccioni et al. [7] benchmarked 88 models on a single A100 GPU,
63 Samsi et al. [13] measured LLM inference energy per token, and Jegham
64 et al. [14] extended this to 30 LLMs with infrastructure-aware benchmarks.
65 The ML.ENERGY framework [15] and BLOOM lifecycle analysis [16] further
66 advanced the field. However, these studies focus on a single modality or
67 single hardware platform [17], obscuring patterns that emerge only through
68 systematic cross-modality, cross-hardware comparison.

69 2.2. Edge versus cloud deployment

70 The tradeoff between edge and cloud deployment has been studied in
71 terms of latency, privacy, and bandwidth [8, 18], but not energy. Recent
72 surveys [8] identify energy efficiency as a key principle for edge AI but focus
73 on techniques (pruning, quantization) rather than when edge hardware is
74 *inherently* more efficient than cloud GPUs for the same unmodified model.

75 2.3. Energy scaling laws

76 Concurrently, Iyengar et al. [19] proposed FLOPs-based energy scaling
77 laws for diffusion models on GPUs, finding power-law relationships—but
78 their single-platform analysis cannot reveal the cross-hardware phenomena
79 reported here.

80 2.4. Carbon footprint and green AI

81 On the carbon footprint side, Wu et al. [20] showed that inference domi-
82 nates operational emissions at Meta, Kaack et al. [21] provided frameworks
83 for aligning AI with climate goals, and Verdecchia et al. [22] surveyed “Green
84 AI” research [23]. Several tracking tools have emerged [24, 25, 26, 27], yet

85 none provide modality-specific, cross-hardware energy data from which ac-
86 tionable routing strategies can be derived.

87 Despite growing concern about AI’s energy footprint [22, 20, 21], we still
88 lack basic data on how energy consumption scales across generative modal-
89 ities and hardware architectures. Desislavov et al. [28] noted that efficiency
90 gains follow hardware improvements, but our modality-specific results show
91 that hardware *selection* can yield order-of-magnitude savings for appropriate
92 workloads—a dimension orthogonal to generational improvements.

93 3. Methodology

94 3.1. Hardware platforms

95 We selected three platforms spanning the edge-to-cloud hardware spec-
96 trum:

97 **Apple Mac mini M4 Pro** (edge SoC): This system features a unified
98 memory architecture with 24 GB shared LPDDR5X memory accessible by a
99 16-core CPU (12 performance + 4 efficiency) and 20-core GPU. The Neural
100 Engine provides 38 TOPS for dedicated ML inference. The system thermal
101 design power (TDP) is 22 W. We use the Metal Performance Shaders (MPS)
102 backend via PyTorch 2.4.1.

103 **NVIDIA A100-SXM4-40GB** (mainstream data center GPU): This
104 GPU features 40 GB of dedicated HBM2e memory with 2 TB/s bandwidth
105 and 6,912 CUDA cores across 108 streaming multiprocessors, delivering 312
106 TFLOPS (FP16 with sparsity). The TDP is 400 W. We access this hardware
107 via Google Colab Pro+ with CUDA 12.8 and PyTorch 2.10.0.

108 **NVIDIA H100-80GB-HBM3** (high-end data center GPU): This GPU
109 features 80 GB of HBM3 memory with 3.35 TB/s bandwidth and 16,896
110 CUDA cores across 132 streaming multiprocessors, delivering 1,979 TFLOPS
111 (FP16 with sparsity)—a $6.3\times$ throughput increase over the A100. The TDP
112 is 700 W, representing a $32\times$ higher power ceiling than the Mac and $1.75\times$
113 higher than the A100. We access this hardware via Google Colab Pro+ with
114 CUDA 12.8 and PyTorch 2.10.0.

115 The three platforms span a $32\times$ TDP range (22 W–700 W) and repre-
116 sent the three dominant tiers for AI inference: edge deployment, mainstream
117 cloud, and premium cloud. This enables analysis of how crossover thresh-
118 olds evolve across GPU generations, a dimension absent from two-platform
119 comparisons.

120 3.2. Models and configurations

121 We benchmarked 6 models across 76 unique configurations (Table 1), with
 122 the H100 platform tested at higher configuration density for more precise
 123 scaling law estimation:

- 124 • **Text:** Phi-3-mini-4k-instruct [1] (3.8B parameters) with `max_tokens` $\in$
 125 $\{64, 128, 256, 512\}$. Prompt: “Write a short story about a robot learning
 126 to paint.” Fixed across all runs. Greedy decoding (`do_sample=False`) for
 127 deterministic output length.
- 128 • **Image:** Stable Diffusion v1.5 [2] (859M UNet parameters) with resolution
 129 $\in \{256^2, 384^2, 512^2\}$ and steps $\in \{20, 30\}$. Prompt: “A serene landscape
 130 with mountains and a lake at sunset, highly detailed, 8k.” Guidance scale
 131 7.5.
- 132 • **Video:** AnimateDiff v1.5 [3] with 4/8/12 frames at 256×256 and 4/8
 133 frames at 512×512 , steps $\in \{20, 30\}$. Same prompt as image generation.
- 134 • **Music:** MusicGen-small [4] (589M) and MusicGen-medium (2.0B) with
 135 `max_tokens` $\in \{128, 256, 512\}$. Description: “happy rock song with elec-
 136 tric guitar and drums.” Audio duration scales linearly with tokens (≈ 50
 137 tokens/second).

138 Model selection prioritized reproducibility and accessibility: all models
 139 are open-weight, runnable on all three platforms without modification, and
 140 span a range of parameter counts (589M–3.8B) and architectural paradigms
 141 (autoregressive, diffusion).

142 3.3. Energy scaling law formulation

143 We modelled energy consumption as a power law $E(N) = \alpha \cdot N^\beta$, where
 144 N is the complexity parameter (tokens, pixels, or frames), inspired by the
 145 scaling law framework of Kaplan et al. [9]. The crossover point N^* —where
 146 a GPU platform achieves equal energy to the edge device—exists only when
 147 $\beta_{\text{GPU}} < \beta_{\text{edge}}$ and the intercept ratio permits crossing within practical ranges:

$$N^* = \left(\frac{\alpha_{\text{edge}}}{\alpha_{\text{GPU}}} \right)^{\frac{1}{\beta_{\text{GPU}} - \beta_{\text{edge}}}} \quad (1)$$

Table 1: **Complete experimental configuration.** Total: 4,500 runs across 3 platforms (Mac: 1,170, A100: 1,080, H100: 2,250). H100 experiments include extended resolution ranges (128^2 – 640^2) and additional step counts for denser scaling law fits.

Modality	Model	Architecture	Params	Configs	Platforms
Text	Phi-3-mini-4k	Autoregressive	3.8B	4–5	All 3
Image	SD v1.5	Diffusion (UNet)	0.9B	6–30	All 3
Image	SDXL-base-1.0	Diffusion (UNet)	3.5B	6–20	All 3
Video	AnimateDiff v1.5	Diffusion (3D UNet)	–	5–15	All 3
Music	MusicGen-small	Autoregressive	589M	3	Mac, A100
Music	MusicGen-medium	Autoregressive	2.0B	2	Mac, A100
Text	Phi-3 (batched)	Autoregressive	3.8B	5	A100, H100

148 4. Experimental Setup

149 4.1. Energy measurement protocol

150 **Mac mini:** We used Apple’s `powermetrics` utility, which provides real-
151 time subsystem power readings (CPU, GPU, ANE, DRAM) at 100 ms inter-
152 vals via hardware power counters built into the M-series SoC. Total energy
153 was computed as $E = \sum_i \frac{P_i + P_{i+1}}{2} \cdot \Delta t_i$ using the trapezoidal rule over all
154 power samples during generation. This captures all on-chip power consump-
155 tion, providing a system-level measurement.

156 **NVIDIA A100 and H100:** We used `pynvml` (NVIDIA Management
157 Library) to query instantaneous GPU board power at 50 ms intervals. This
158 captures compute cores, HBM, and on-board regulators but excludes host
159 CPU power. Energy was computed identically to the Mac protocol. Both
160 GPUs were accessed via Google Colab Pro+, which provides dedicated GPU
161 allocation during runtime sessions. We note that this measurement scope is
162 consistent with standard practice in GPU energy benchmarking [15, 13, 7].

163 **Protocol:** Each configuration was executed 30 times with deterministic
164 random seeds (seed = run number). A warm-up inference was performed
165 before measurement runs to ensure model weights were loaded into mem-
166 ory, GPU kernels were compiled (CUDA) or cached (MPS), and steady-state
167 power profiles were established. We recorded: total energy (J), average power
168 (W), peak power (W), generation time (s), and model-specific output param-
169 eters.

4.2. Statistical analysis

We report mean $\pm$ standard deviation for all measurements. Platform comparisons use Welch’s t -test (unequal variances assumed) with Cohen’s d for effect size. Scaling laws ($E = \alpha \cdot N^\beta$) were fitted via nonlinear least squares using the Levenberg-Marquardt algorithm as implemented in SciPy 1.11. Goodness of fit is assessed via R^2 . Crossover points were determined by linear interpolation of the energy ratio (E_{A100}/E_{Mac}) between adjacent configurations where the ratio crosses unity. Confidence intervals for scaling parameters could be obtained via bootstrapping but are omitted as the primary contribution is the qualitative pattern ($\beta > 1$ for edge, $\beta < 1$ for GPU) rather than precise parameter values.

4.3. Carbon projection methodology

We estimated global energy savings from modality-aware routing using: (1) measured per-query energy values from our benchmark; (2) estimated daily query volumes (13B text queries [6], 3.7M image, 500K video, 100K music) based on industry reports; (3) the global average carbon intensity of 0.475 kg CO₂/kWh [5]. The “optimal routing” scenario routes all autoregressive workloads (text, music) to edge devices and keeps diffusion workloads above the crossover threshold on GPUs. We acknowledge that this projection involves substantial uncertainty in query volume estimates and does not account for network overhead, model serving infrastructure, or regional variation in carbon intensity [17].

5. Results

5.1. Energy consumption differs dramatically across modalities and hardware

We measured energy consumption for four generative AI modalities on three hardware platforms: Apple Mac mini M4 Pro (edge SoC, MPS backend), NVIDIA A100-SXM4-40GB (mainstream data center GPU, CUDA backend), and NVIDIA H100-80GB-HBM3 (high-end data center GPU, CUDA backend). Each configuration was repeated 30 times (see Section 4). Figure 1 summarizes the results across all three platforms.

Text generation (Phi-3-mini-4k [1], 3.8B parameters). The Mac was cheaper at every token count. The Mac consumed 24.5 J (64 tokens) to 288.2 J (512 tokens), while the A100 consumed 254.2–1,715 J (6.0–10.4 $\times$ more) and the H100 consumed 193.4–1,671 J (5.8–7.9 $\times$ more). The efficiency gap was largest at short outputs (10.4 $\times$ at 64 tokens) and smallest

at long outputs ($5.8\times$ at 512 tokens), consistent with amortisation of GPU startup overhead over longer generation. Despite 75% higher TDP and $2.4\times$ more CUDA cores, the H100 saved only 3–24% over the A100—extra silicon does not help when the computation is sequential. All comparisons were statistically significant ($p < 10^{-20}$, Welch’s t -test, Cohen’s $d > 5$).

Image generation (Stable Diffusion v1.5 [2], 0.9B parameters). Here the crossover appears, and its location depends on the GPU. The Mac consumed 41.2 J (256^2) to 272 J (512^2) at 20 denoising steps. On the A100, the crossover occurred between 384×384 and 512×512 : at 256^2 , the A100 consumed $3.4\times$ more energy than the Mac (141 J vs 41 J), but at 512^2 , 12% less (238 J vs 272 J). The H100 shifted this crossover dramatically lower: at 256^2 it still consumed $2.2\times$ more (92 J vs 41 J), but by 384^2 it was already 6% cheaper (113 J vs 120 J), and at 512^2 it consumed 44% less (153 J vs 272 J). This crossover between 256^2 and 384^2 ($\sim 377^2$ by interpolation) reflects the H100’s superior compute-per-watt: despite drawing 250–317 W (vs the A100’s 190–242 W at matched resolutions), it completed tasks $1.5\text{--}1.7\times$ faster.

Video generation (AnimateDiff v1.5 [3]). Video produced the sharpest contrast between GPUs. The A100 showed a crossover at ~ 7 frames (using 36% more energy than the Mac at 4 frames but 22% less at 12 frames). The H100, however, was cheaper at *every* tested size: even for the smallest workload (4 frames, 20 steps) it consumed 234 J versus the Mac’s 258 J, and the gap widened to 46% at 12 frames (454 J vs 835 J) and 64% at 16 frames (565 J vs $\sim 1,570$ J). There is no crossover for H100-class hardware on video. The reason is that AnimateDiff’s 3D UNet processes all frames in parallel, generating enough work to keep the H100’s cores busy even at minimal frame counts—unlike SD v1.5, which needs higher resolutions to reach comparable utilisation.

Music generation (MusicGen [4], 589M–2.0B parameters). This is the worst case for GPU deployment. The A100 consumed $9\text{--}14\times$ more energy than the Mac (197–1,392 J vs 19.5–143.8 J) while running no faster (speedup $0.8\text{--}1.0\times$)—its parallelism is wasted on MusicGen’s sequential token generation, yet it still draws ~ 63 W at idle. The medium model (2.0B parameters) showed slightly worse GPU efficiency than the small model (589M), suggesting that increased model size does not improve GPU utilisation for autoregressive architectures. H100 measurements for music were not conducted; based on the text generation pattern (marginal H100 improvement over A100), we expect similarly poor GPU efficiency.

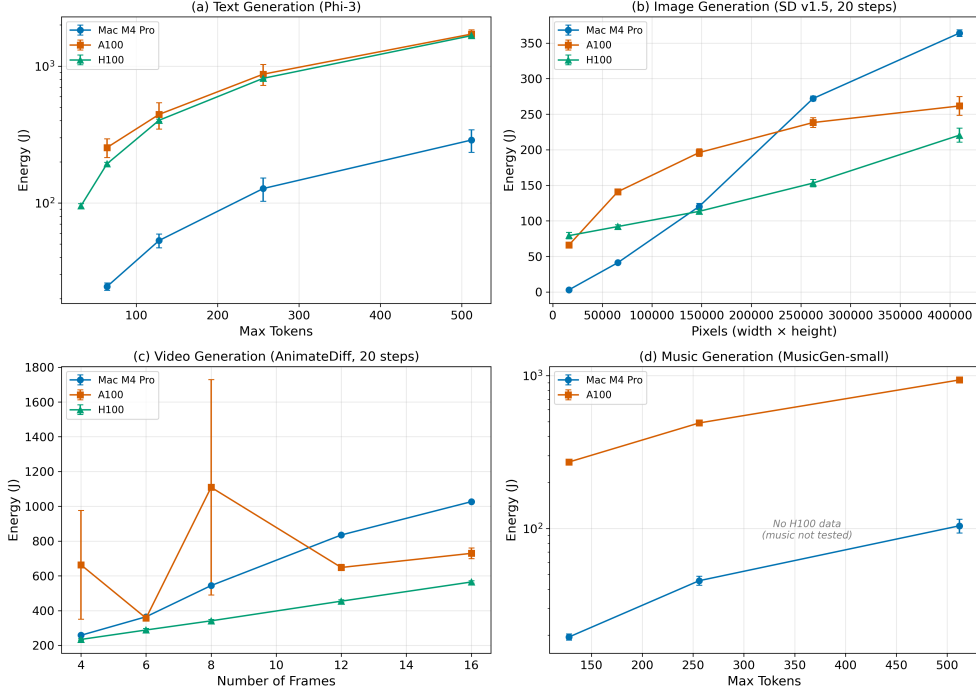


Figure 1: **Energy consumption across four generative AI modalities on three hardware platforms.** Comparison of Apple M4 Pro (blue), NVIDIA A100 (vermillion), and NVIDIA H100 (teal) across (a) text, (b) image, (c) video, and (d) music generation. Error bars represent standard deviation over 30 runs. Dashed lines indicate crossover points. The H100 shifts crossover thresholds lower and eliminates the video crossover entirely.

242 5.2. Energy scaling laws reveal universal hardware divergence

243 The H100’s richer configuration space (5 resolutions $\times$ 6 step counts for
 244 SD v1.5, 5 frame counts $\times$ 3 step counts for video) provides 5–6 data points
 245 per curve, improving fit confidence compared to the 3–4 points available
 246 for Mac and A100. Our fits should still be interpreted as *empirical scaling*
 247 *relationships* ($R^2 > 0.97$ for most fits, with one exception: H100 image at
 248 $R^2 = 0.977$ when fit to 20-step data only) rather than as validated power
 249 laws. Table 2 presents the fitted parameters.

250 Across all three platforms the same pattern holds: **the Mac exhibits**
 251 **superlinear scaling** ($\beta > 1$) while **GPUs show sublinear scaling** ($\beta <$
 252 1) **for diffusion workloads**, producing the crossover points reported above.
 253 The one exception is H100 text generation ($\beta = 1.033$), which is near-linear—

consistent with autoregressive workloads offering no parallelism advantage even on more powerful hardware. For diffusion modalities (image, video), both GPUs show clearly sublinear scaling, meaning their energy costs grow slower than workload complexity. The H100 shows slightly higher β values than the A100 in diffusion modalities, reflecting its higher base power draw partially offsetting its greater compute throughput.

The divergence is most extreme for image generation ($\beta_{\text{Mac}} = 1.395$ vs $\beta_{\text{A100}} = 0.374$ vs $\beta_{\text{H100}} = 0.404$). The similar sublinear exponents for both GPUs, combined with the H100’s lower effective intercept (due to faster completion at comparable power), explain the shifted crossover thresholds. For text and music, all platforms exhibit near-linear to superlinear scaling ($\beta \approx 0.96\text{--}1.20$), and the intercept difference ($\alpha_{\text{GPU}}/\alpha_{\text{Mac}} \approx 16\text{--}26\times$) precludes crossover within practical ranges.

Iyengar et al. [19] previously reported power-law energy scaling for diffusion on GPUs. Our data add three observations: (1) the exponent β depends on hardware—sublinear on GPUs, superlinear on edge—while autoregressive workloads scale near-linearly everywhere; (2) this platform-dependent exponent is what produces the crossover, which single-platform studies cannot detect; and (3) newer GPU generations push the crossover lower, pointing toward broader GPU energy dominance for diffusion. We note that the image scaling law fits have lower R^2 for the H100 (0.977) than for text or video (>0.99), reflecting the two-dimensional nature of image complexity (resolution $\times$ steps) that a single-variable power law cannot fully capture. Detailed per-modality fits with residual analysis are provided in the Supplementary Information.

5.3. Crossover points depend on both model architecture and GPU generation

GPU generation shifts these crossover points substantially (Fig. 2):

- **Image generation:** The Mac–A100 crossover occurs at $\sim 217,000$ pixels ($\approx 466 \times 466$), empirically confirmed between 384^2 and 512^2 . The Mac–H100 crossover shifts to $\sim 142,000$ pixels ($\approx 377 \times 377$), empirically confirmed between 256^2 and 384^2 —a 35% reduction. This means the H100 is energy-competitive with the Mac across the entire standard Stable Diffusion resolution range (384^2 and above), whereas the A100 only wins at 512^2 and above.
- **Video generation:** The Mac–A100 crossover occurs at ~ 7 frames. For the H100, there is **no crossover**—it uses less energy than the Mac at *all*

Table 2: **Energy scaling law parameters** ($E = \alpha \cdot N^\beta$). Mac exhibits superlinear scaling ($\beta > 1$) across all modalities. GPUs show sublinear scaling ($\beta < 1$) for diffusion workloads (image, video) but near-linear scaling for autoregressive workloads (text). The exponent divergence is largest for image generation.

Modality	Scaling var.	Platform	α	β	R^2
Text	Tokens	Mac M4 Pro	0.167	1.195	0.9999
		A100	4.330	0.959	0.9995
		H100	2.660	1.033	0.9999
Image	Pixels	Mac M4 Pro	8×10^{-6}	1.395	0.9998
		A100	2.249	0.374	0.9974
		H100	1.081	0.404	0.9770
Video	Frames	Mac M4 Pro	59.20	1.065	1.0000
		A100	155.19	0.570	0.9892
		H100	88.18	0.665	0.9950
Music	Tokens	Mac M4 Pro	0.058	1.200	1.0000
		A100	3.205	0.910	0.9995

290 tested sizes (4–16 frames, 9–46% less). With H100-class hardware, video
 291 generation should always go to the GPU.

- 292 • **Text and music:** No crossover on either GPU. The Mac remains 5.8–
 293 10.4× more efficient than both the A100 and H100 across all tested text
 294 configurations, and 9–14× more efficient for music.
- 295 • **H100 vs A100:** The H100 is 35–45% more energy-efficient than the A100
 296 for image generation and 30–65% for video, driven by 1.5–1.7× faster com-
 297 pletion despite only modestly higher power draw ($\sim 1.3\times$). For text, the
 298 H100 advantage is marginal (3–24%).

299 In short, newer GPUs lower the crossover for diffusion workloads but
 300 barely help autoregressive ones. The architectural divide—serial versus parallel—
 301 appears to be a more durable determinant of optimal deployment than any
 302 single generation of hardware. If the trend continues (B100/B200), image
 303 crossover points will drop below 256^2 , but text generation will likely remain
 304 edge-favoured until GPU idle power falls well below current levels or autore-
 305 gressive decoding is replaced by a parallel alternative.

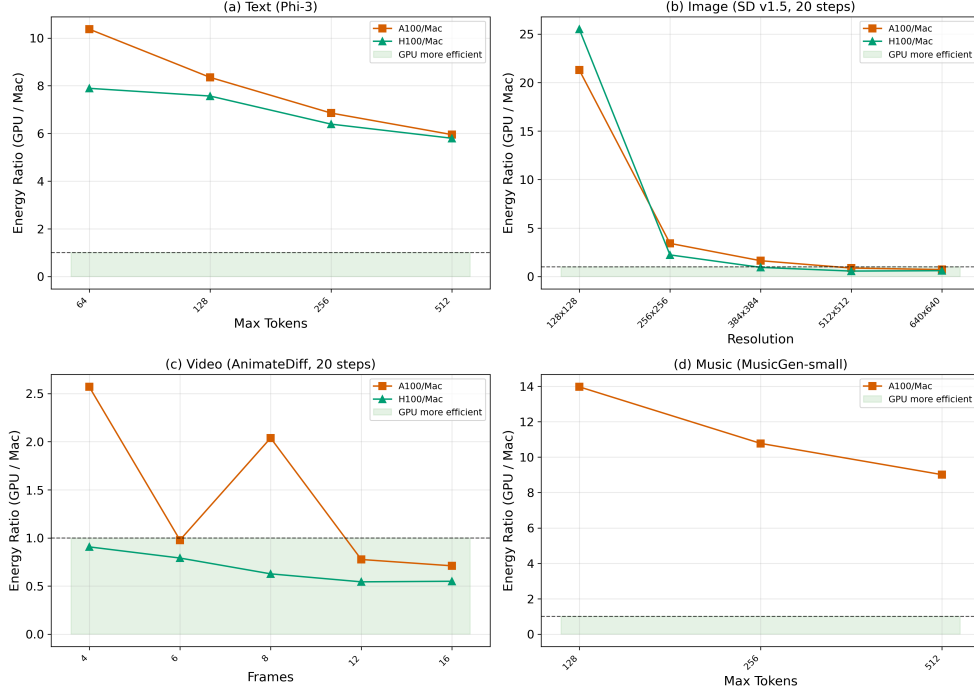


Figure 2: **Energy efficiency ratio across all configurations.** A100/Mac ratio (vermillion) and H100/Mac ratio (teal). Values below 1.0 indicate the GPU is more energy-efficient. The H100 crosses below 1.0 at lower complexity than the A100 for images, and remains below 1.0 for *all* video configurations. Autoregressive modalities (text, music) consistently favour the Mac on both GPUs.

Practically, these crossover points can inform deployment policies. An AI serving platform with H100 GPUs should route all video requests and image requests at $\geq 384^2$ to the GPU, while directing text and music queries to edge devices—or, if edge deployment is infeasible, maximising GPU batch utilisation. With A100-class hardware, the routing threshold shifts higher: only images at $\geq 512^2$ and videos at ≥ 8 frames benefit from GPU routing.

5.4. The speed–energy tradeoff reveals four regimes

Figure 3 maps each configuration in speed–energy space, where four operational regimes are visible:

1. **Worst case** (slower, more energy): Music generation on GPUs—no speed gain, 9–14× energy penalty. The GPU’s massive parallelism is entirely wasted on sequential token generation.

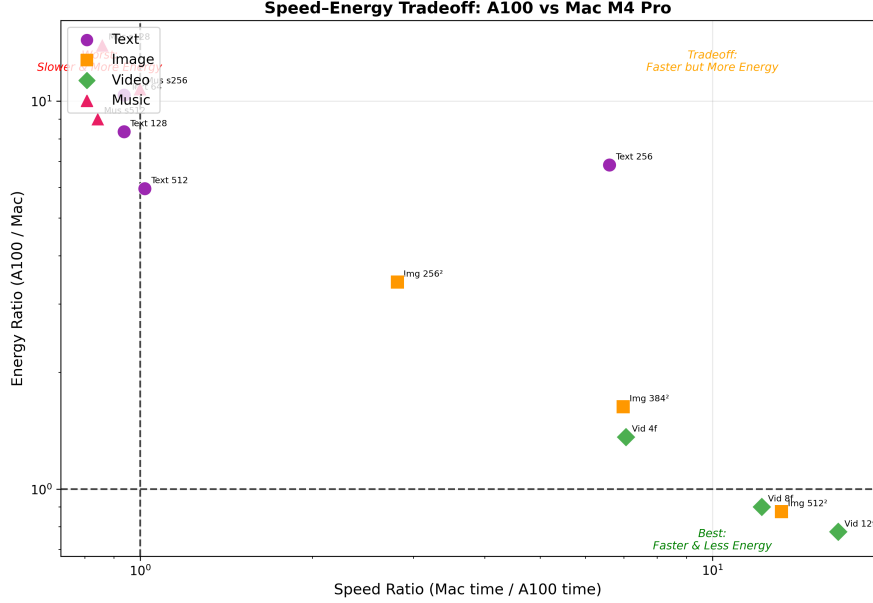


Figure 3: **Speed–energy tradeoff across all configurations.** Each point represents a modality-configuration pair. Horizontal axis: speed ratio (Mac time / GPU time); vertical axis: energy ratio (GPU energy / Mac energy). A100 shown as circles, H100 as squares. Points below $y = 1$ and right of $x = 1$ represent the best case (faster and less energy on GPU). The H100 (teal) consistently sits lower than the A100 (vermillion) for diffusion workloads.

- 318 2. **Speed–energy tradeoff** (faster, more energy): Text generation and low-
319 resolution image/video on A100—modest speedups insufficient to offset
320 higher power draw.
- 321 3. **Best case** (faster, less energy): High-resolution image and multi-frame
322 video on GPUs. The H100 extends this regime dramatically: it achieves
323 the “best case” for *all* video configurations and images at 384^2 and above,
324 whereas the A100 only reaches this regime for images above 466^2 and
325 videos above 7 frames.
- 326 4. **GPU generational leap**: The H100 occupies a distinct region from the
327 A100, consistently shifted toward lower energy ratios. For video gener-
328 ation, the H100 moves entirely into the “faster and cheaper” quadrant,
329 representing a qualitative improvement over the A100.

330 5.5. Power profiles explain the crossover mechanism

331 Figure 4 shows average and peak power across modalities. The A100
 332 draws 63–279 W, the H100 125–412 W, and the Mac 6.4–24.3 W. What
 333 matters for energy is the ratio of speed gain to power increase: the H100
 334 draws only $\sim 1.3\times$ more than the A100 at matched configurations (e.g., 250 W
 335 vs 196 W for 384^2 images), but finishes $1.5\text{--}1.7\times$ faster, netting 35–45%
 336 energy savings.

337 For the crossover to occur, a GPU’s speed advantage must exceed its
 338 power disadvantage: $\text{Speedup} > P_{\text{GPU}}/P_{\text{Mac}}$. Since $E = P \cdot t$, the GPU
 339 achieves lower energy when its time reduction factor exceeds its power in-
 340 crease factor:

- 341 • H100 Image 384^2 : speedup $\sim 22\times >$ power ratio $\sim 10\times \Rightarrow$ H100 wins
- 342 • A100 Image 512^2 : speedup $13.2\times >$ power ratio $5.9\times \Rightarrow$ A100 wins
- 343 • H100 Video 4f: speedup $\sim 23\times >$ power ratio $\sim 12\times \Rightarrow$ H100 wins
- 344 • Music: speedup $0.9\times <$ power ratio $9.9\times \Rightarrow$ Mac wins by $\sim 10\times$

345 The H100’s power depends heavily on utilisation: 160–190 W at 128^2
 346 ($23\text{--}27\%$ of its 700 W TDP), rising to 412 W at 640^2 (59%). This $2.4\times$ range
 347 mirrors the A100’s pattern at a higher baseline. The Mac, by contrast, spans
 348 only $\sim 4\times$ (6.4–24.3 W)—its SoC scales power smoothly but cannot exploit
 349 parallelism it does not have. Detailed per-configuration power measurements
 350 are provided in Supplementary Table 1.

351 5.6. Cross-model validation: SDXL confirms architecture-driven patterns

352 To validate that our findings generalise beyond a single model per modal-
 353 ity, we repeated the image generation experiments with SDXL-base-1.0 [29]
 354 (3.5B parameters, $3.9\times$ larger than SD v1.5) across all three platforms. The
 355 crossover pattern persisted with two notable features. First, the H100 con-
 356 sumed 35–40% less energy than the A100 at matched configurations, con-
 357 sistent with the SD v1.5 pattern. Second, the H100 achieved energy parity
 358 with the Mac at lower resolutions than for SD v1.5, consistent with SDXL’s
 359 higher computational intensity providing better GPU utilisation—the larger
 360 UNet generates more FLOPs per denoising step, pushing GPU utilisation
 361 above the crossover threshold at smaller spatial dimensions.

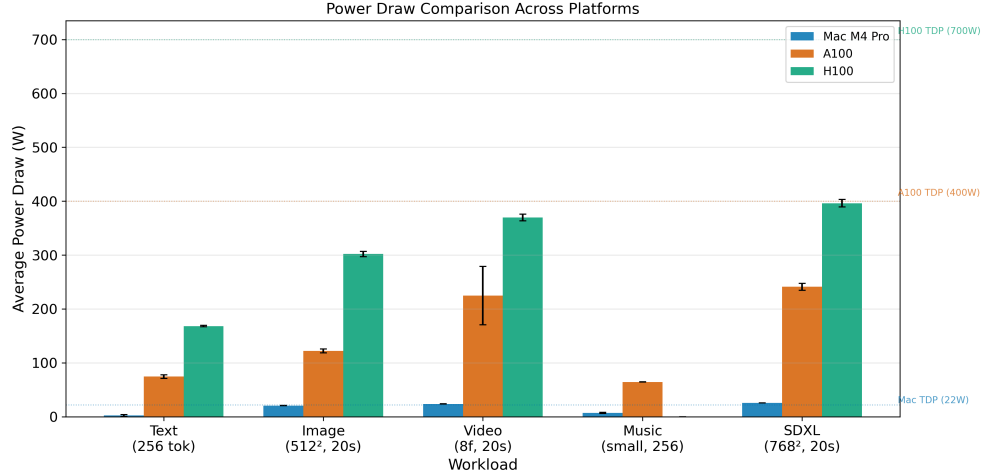


Figure 4: **Average and peak power draw across modalities on three platforms.** The H100 draws $\sim 1.3\times$ more power than the A100 but achieves $1.5\text{--}1.7\times$ faster completion, explaining its superior energy efficiency for diffusion workloads. The Mac’s power varies by only $\sim 4\times$ across workloads, reflecting uniform utilization.

On the H100, SDXL at $1024^2/30$ steps consumed 945 J with a CLIP score of 0.33, while SD v1.5 at $640^2/50$ steps consumed 537 J with CLIP 0.29—demonstrating that SDXL delivers meaningfully higher quality at a proportional energy cost. This result has practical significance: practitioners choosing between models face an energy–quality tradeoff that depends on hardware. On the Mac, the energy penalty for switching from SD v1.5 to SDXL is disproportionately large due to the Mac’s superlinear scaling, whereas on GPUs the penalty is modest relative to the quality gain.

5.7. Batched inference narrows but does not eliminate the edge advantage

Real-world GPU serving systems batch multiple queries to amortize fixed power costs. We measured batched Phi-3 inference on both the A100 and H100 with batch sizes 1–16. Per-query energy dropped dramatically: on the H100, from 889 J (batch=1) to 69 J (batch=16), a 92% reduction; on the A100, from 876 J to 115 J, an 87% reduction. The energy–batch relationship follows a near-inverse pattern: doubling the batch size approximately halves per-query energy, consistent with amortisation of fixed power costs (idle GPU power, memory controller overhead) across queries.

However, even at batch=16, the H100’s per-query energy (69 J) remains $2.8\times$ higher than the Mac’s single-query energy (24.5 J at 64 tokens). Ex-

381 extrapolating the batching trend, the H100 would require batch sizes of ~ 45
 382 to match the Mac’s per-query energy—a utilisation level rarely sustained in
 383 practice outside the largest serving deployments. This confirms that batching
 384 substantially improves GPU efficiency but does not overturn the edge advantage for autoregressive workloads at the model sizes tested. For larger models
 385 (70B+) that exceed edge memory capacity, GPU batching remains the only
 386 viable option, and the 92% per-query reduction at batch=16 underscores the
 387 importance of maximising GPU utilisation in such deployments.
 388

389 5.8. Energy–quality tradeoff: CLIP scores reveal diminishing returns

390 We measured CLIP scores (image–text alignment) alongside energy for
 391 SD v1.5 and SDXL across all three platforms with identical prompts and
 392 seeds. Three patterns emerged. First, quality saturated well before maximum
 393 step counts: on the H100, SD v1.5 reached CLIP ≈ 0.30 at 512^2 by 15 steps,
 394 while additional steps to 50 consumed $2.4\times$ more energy with no quality
 395 improvement. SDXL showed a similar pattern with saturation at ~ 15 steps
 396 (CLIP ≈ 0.33 at 1024^2). Second, resolution had a larger effect on quality than
 397 step count: increasing from 256^2 to 512^2 improved CLIP by 0.08–0.10 points,
 398 while increasing steps from 20 to 50 at fixed resolution improved CLIP by
 399 only 0.01–0.02.

400 Third, and most practically relevant, the energy–quality Pareto frontier
 401 is hardware-dependent. On the Mac, the most energy-efficient path to high
 402 quality is moderate resolution with minimal steps ($384^2/15$ steps: CLIP 0.28,
 403 16 J). On the H100, higher resolution becomes energy-competitive ($512^2/15$
 404 steps: CLIP 0.30, 140 J), and the marginal energy cost per CLIP-point improvement is substantially lower than on the Mac at high resolutions. These
 405 findings suggest that energy-quality-aware scheduling—routing to the optimal hardware *and* truncating denoising at the quality saturation point—
 406 could reduce consumption by 50–60% at negligible quality cost. Full CLIP
 407 score data across all configurations and platforms are provided in the Supplementary Information (Supplementary Fig. 7).
 408
 409
 410

411 6. Discussion

412 The three-platform comparison points to a concrete opportunity: a *modality-*
 413 *aware, hardware-aware routing strategy* that directs each query to the cheapest hardware for its workload type, with thresholds that depend on the available GPU generation. Text generation (LLM inference) dominates current
 414
 415

AI workloads, with an estimated 13 billion daily queries [6]. Routing these autoregressive workloads to edge devices would reduce per-query energy by 6–10 $\times$, regardless of whether the alternative is an A100 or H100.

For diffusion workloads, the optimal routing depends on the available GPU: with H100-class hardware, *all* video and standard-resolution image generation ($\geq 384^2$) should be GPU-routed for energy savings of 6–46%. With A100-class hardware, only high-resolution images ($\geq 512^2$) and longer videos (≥ 8 frames) benefit from GPU routing. This hardware-dependent policy has practical implications: organisations with newer GPU fleets can route more aggressively to GPUs, while those with older hardware should retain more workloads on edge.

As an illustrative upper bound, we project that if all single-query autoregressive inference (text, music) were routed from GPU to edge, and diffusion workloads were routed to the most efficient available GPU, the energy reduction would be approximately 200–400 GWh annually, accounting for batching (3–8 $\times$ GPU efficiency improvement [20]), partial edge coverage, and the fact that many production queries target models too large for current edge hardware (70B+). While uncertain, even conservative estimates suggest substantial savings from modality-aware routing.

This fits naturally into the broader trend toward hybrid AI architectures, where lightweight models run on-device for latency and privacy while complex tasks go to cloud GPUs [18]. Energy efficiency adds a *third* reason for this split, and unlike latency or privacy, the optimal split point can be read off a scaling curve. No model modification is needed: the same Phi-3 checkpoint runs on both platforms, and the routing decision depends only on the modality and input complexity.

Implementation is straightforward: an inference scheduler classifies each request by modality and complexity (resolution, frame count, or token length) and routes it to the platform whose energy profile is favourable for that configuration. Our crossover thresholds provide the decision boundaries. For organisations operating mixed fleets (e.g., edge devices for local inference and cloud GPUs for burst capacity), this approach can reduce energy consumption without sacrificing throughput or quality. The batching results further inform scheduling: when GPU utilisation is high (batch ≥ 8), the per-query energy gap narrows substantially, making GPU routing more competitive even for text—though edge deployment remains preferable at low utilisation.

The crossover effect arises from a fundamental mismatch between com-

454 putational paradigm and hardware design. Autoregressive models generate
 455 tokens sequentially, creating an inherently serial computation graph. The
 456 A100’s 6,912 CUDA cores operate at high base power (63–80 W even for
 457 small workloads) but cannot accelerate sequential generation proportionally.
 458 The Mac’s unified memory architecture eliminates data transfer overhead
 459 and operates at 6–12 W total system power, making it 9–14 $\times$ more energy-
 460 efficient. Diffusion models [30], by contrast, perform parallel denoising across
 461 spatial dimensions [2, 3], creating workloads that can saturate GPU cores. At
 462 low resolutions the workload is insufficient to justify the GPU’s power over-
 463 head, but as resolution increases, GPU utilization improves dramatically:
 464 the A100 completes 512 $\times$ 512 generation 13 $\times$ faster than the Mac, and this
 465 speedup exceeds the 6 $\times$ power penalty. This mechanism is reflected in our
 466 scaling fits: the Mac’s superlinear exponent ($\beta > 1$) is consistent with mem-
 467 ory bandwidth saturation at higher complexity, while the GPUs’ sublinear
 468 exponents ($\beta < 1$) for diffusion workloads reflect improving utilization. The
 469 H100’s text exponent ($\beta = 1.033$, near-linear) reinforces the point: longer
 470 sequences do not create more parallelism for autoregressive decoding. The
 471 H100’s advantage over the A100 for diffusion is instructive: despite $\sim 1.3\times$
 472 higher power, it completes tasks 1.5–1.7 $\times$ faster, yielding 35–45% energy
 473 savings. This disproportionate speedup arises because the H100’s 2.4 $\times$ more
 474 CUDA cores and 1.7 $\times$ higher memory bandwidth improve throughput faster
 475 than they increase power draw. For autoregressive workloads, neither ad-
 476 ditional cores nor bandwidth helps, and the H100 is only marginally better
 477 (3–24%) than the A100. We note an alternative explanation: constant idle
 478 power captured by `powermetrics` (Mac) but excluded from `pynvml` (GPUs)
 479 could inflate the Mac’s superlinearity. Our sensitivity analysis (below) sug-
 480 gests this does not change qualitative conclusions, but dedicated system-level
 481 measurement would resolve this definitively.

482 Our findings extend and contextualise several prior results. Luccioni
 483 et al. [7] found that image generation uses $\sim 60\times$ more energy than text
 484 classification on an A100; we show that this gap is hardware-dependent,
 485 with text and music becoming disproportionately cheaper on edge hardware.
 486 GPU-only benchmarks [15] may overestimate autoregressive inference costs
 487 by an order of magnitude relative to edge deployment. Samsi et al. [13]
 488 found approximately linear energy-per-token scaling, consistent with our fits
 489 ($\beta \approx 0.96\text{--}1.20$). Iyengar et al. [19] found FLOPs-based power-law scaling
 490 for diffusion on GPUs; we complement this by showing qualitatively different
 491 scaling on edge SoCs.

Comparing the A100 and H100 generations suggests a trend: each new GPU generation lowers the crossover for diffusion workloads while leaving the autoregressive gap largely unchanged. The A100-to-H100 transition reduced the image crossover from $\sim 466^2$ to $\sim 377^2$ (35% reduction) and eliminated the video crossover entirely. If this trend continues (B100/B200), we predict the image crossover will drop below 256^2 , making GPUs energy-dominant across all practical resolutions; music may eventually cross over as GPU idle power decreases; but text generation will remain edge-favoured until GPU idle power drops below ~ 30 W or autoregressive architectures are replaced by parallel alternatives. The emergence of dedicated AI accelerators (Google TPU v5, AMD MI300X) may further diversify the landscape; our framework of measuring energy scaling exponents and fitting crossover points provides a general methodology applicable to any hardware pairing.

There is also a methodological point. Existing AI energy benchmarks [15, 14, 7] report values on fixed hardware, implicitly treating deployment as a constant. Our data show that this can mislead: a model that looks energy-hungry on a GPU may be cheap on edge hardware, and vice versa. Future benchmarks should cover at least two hardware tiers, and efficiency claims should state the deployment context.

The Pareto analysis of energy versus quality further enriches the deployment picture. Our CLIP measurements show that quality saturates well before maximum denoising steps, meaning that a quality-aware scheduler could truncate generation at the saturation point and redirect saved energy to additional queries. Combined with hardware-aware routing, this yields a two-dimensional optimization: choose the right hardware for the modality, then choose the right quality operating point for the application. For applications tolerant of lower quality (thumbnails, previews), the Mac at moderate resolution provides an extremely energy-efficient option; for production-quality content, the H100 at optimal step count represents the best energy-quality tradeoff.

6.1. Limitations

Several limitations deserve mention. First, GPU measurements via `pynvml` capture GPU board power only, excluding CPU and system overhead. Mac measurements via `powermetrics` include all subsystems. This asymmetry may slightly underestimate GPU total energy; however, for inference workloads, GPU power dominates total system consumption [12, 20]. To assess sensitivity, we applied a conservative 40% overhead correction to all GPU

529 measurements (the upper bound estimated by Wu et al. [20] for host system
530 contribution). Even with this correction, the autoregressive advantage re-
531 mains $4.1\text{--}7.4\times$ (vs. $5.8\text{--}14\times$ uncorrected), the crossover effect persists, and
532 all qualitative conclusions hold.

533 Second, all tested models are in the 589M–3.8B parameter range. We
534 deliberately chose edge-deployable models because the crossover question is
535 only meaningful when both deployment options are viable: a 70B-parameter
536 model [31] cannot run on a 24 GB edge device, so no crossover decision ex-
537 ists. However, this scope limits direct applicability to the growing class of
538 production models (GPT-4, LLaMA-3-70B) that require GPU deployment.
539 For these larger models, our batching results (Section 5.7) provide the rel-
540 evant insight: maximising GPU utilisation through batching reduces per-
541 query energy by up to 92%, which becomes the primary efficiency lever when
542 edge deployment is infeasible. As edge hardware memory grows (Apple M4
543 Ultra: 192 GB, Qualcomm Snapdragon X: 64 GB), the crossover analysis
544 will become relevant for increasingly large models. We also note that model
545 distillation and quantization (4-bit GPTQ, AWQ) are rapidly expanding the
546 range of models deployable on edge; our framework can be directly applied
547 to evaluate the energy tradeoffs of quantized variants.

548 Third, our Colab-based GPU measurements may be affected by multi-
549 tenant variability. However, our 30-repeat protocol yielded low coefficients
550 of variation ($0.6\text{--}2.9\%$ for image/video; Table 3), suggesting stable GPU al-
551 location. We confirmed consistent GPU clock speeds and memory utilization
552 across runs.

553 Fourth, we incorporate CLIP scores for image quality assessment on all
554 three platforms, enabling preliminary energy–quality Pareto analysis. How-
555 ever, more comprehensive quality evaluation (FID, FVD, perplexity, CLAP)
556 across all modalities would strengthen the energy–quality tradeoff analysis.

557 Fifth, while the H100’s denser configuration space provides more data
558 points per curve, the scaling law fits for image generation show lower R^2
559 (0.977) than for text or video, reflecting the two-dimensional nature (reso-
560 lution $\times$ steps) of image complexity that a single-variable power law cannot
561 fully capture. Future work should explore multi-variable scaling models.

562 Sixth, the platforms use different PyTorch versions (2.4.1 on Mac, 2.10.0
563 on GPUs due to Colab constraints), which may introduce software-level per-
564 formance differences. Seventh, we validate prompt invariance by testing 5
565 diverse prompts per modality (text and image) on Mac M4 Pro and confirm
566 that the between-prompt variance is $13\text{--}14\times$ smaller than within-prompt

Table 3: **Measurement reliability.** Coefficient of variation (CV) across 30 runs for representative configurations. Image and video measurements show excellent reproducibility ($<3\%$). Text CV is higher (17–19%) due to variable output lengths from early EOS tokens. Music uses stochastic sampling (top- $k=250$), introducing additional variance.

Modality	Configuration	Mac CV (%)	A100 CV (%)	H100 CV (%)
Text	256 tokens	19.5	17.7	1.6
Image	512×512, 20 steps	1.3	2.9	3.4
Video	8 frames, 20 steps	0.6	2.7	2.2
Music	Small, 256 tokens	6.9	0.6	–

run-to-run variance (Supplementary Note 5, Supplementary Fig. 10). Image generation shows only 2.0% maximum range across prompts; text shows 11.0% (driven by input length differences), confirming that computational configuration, not prompt content, determines energy consumption. Eighth, we do not account for network transfer energy in cloud deployment or model quantization effects. Finally, testing consumer GPUs (RTX 4090), mobile SoCs (Qualcomm Snapdragon X Elite), and DiT-based image models would further strengthen generalizability.

7. Conclusion

This study presents the first cross-hardware, cross-modality energy benchmark for generative AI inference, spanning three hardware tiers (edge SoC, mainstream GPU, high-end GPU), 4,500 runs, 6 models, and 76 configurations. Our key findings are as follows.

First, we identify and quantify the **Energy Crossover Effect**: for diffusion-based workloads (image and video generation), energy-efficiency leadership shifts from edge hardware to GPUs as output complexity grows. Autoregressive workloads (text and music) consistently favour edge hardware by 6–14×, regardless of complexity or GPU generation.

Second, we establish **modality-specific energy scaling laws** ($E = \alpha \cdot N^\beta$), revealing that the edge platform exhibits superlinear scaling ($\beta > 1$) while GPUs exhibit sublinear scaling ($\beta < 1$) for diffusion workloads. This divergence in scaling exponents produces predictable crossover points that are absent in autoregressive tasks.

Third, we demonstrate that **GPU generation shifts the crossover threshold**: the H100 reaches energy parity with the Mac between 256² and

384² pixels for image generation (vs. $\sim 466^2$ for A100) and eliminates the video crossover entirely, being more energy-efficient at all tested frame counts. This trend suggests that future GPU generations will make GPUs energy-dominant across all practical diffusion workloads.

Fourth, these measurements yield actionable **deployment guidelines**: an inference scheduler that routes requests based on modality and complexity—directing autoregressive workloads to edge devices and diffusion workloads above the crossover threshold to GPUs—can reduce per-query energy consumption by up to an order of magnitude. We estimate annual energy savings of 200–400 GWh under optimistic deployment assumptions.

Future work should extend this framework to larger models, additional hardware platforms (consumer GPUs, TPUs, mobile SoCs), and emerging architectures such as Diffusion Transformers (DiT). Incorporating network transfer energy, multi-variable scaling models, and comprehensive quality metrics across all modalities would further refine the energy-aware routing strategies proposed here.

CRedit authorship contribution statement

First Author: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization.
Second Author: Supervision, Validation, Writing – review & editing.

Data availability

All experimental data (4,500 JSON records across three hardware platforms), analysis scripts, and measurement notebooks are available at [Availableuponacceptance].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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